Binary Search Tree

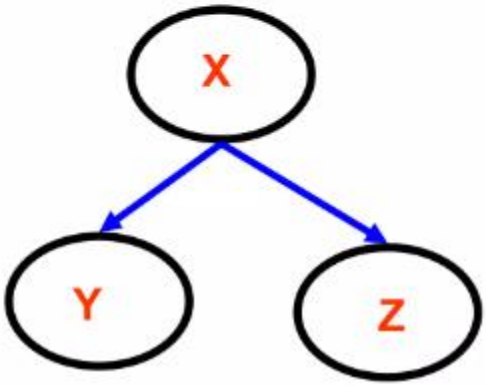
Key property is value at node

- Smaller values in left subtree
- Larger values in right subtree

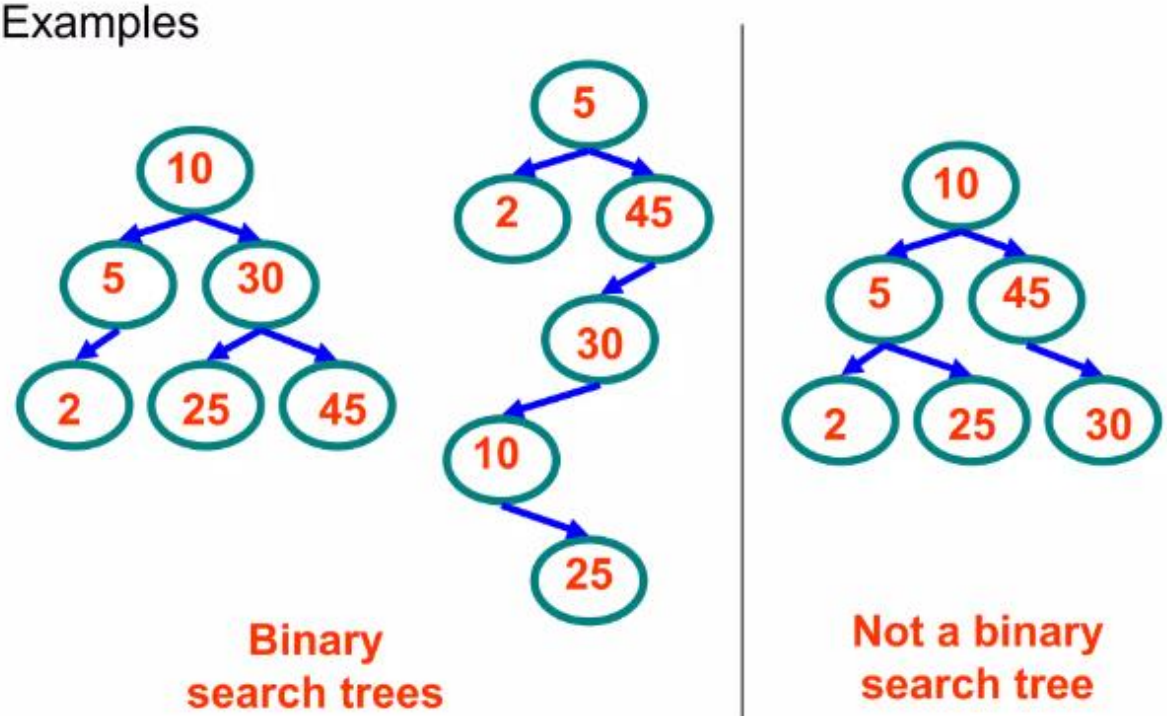
Example

X > Y

X < Z



Examples

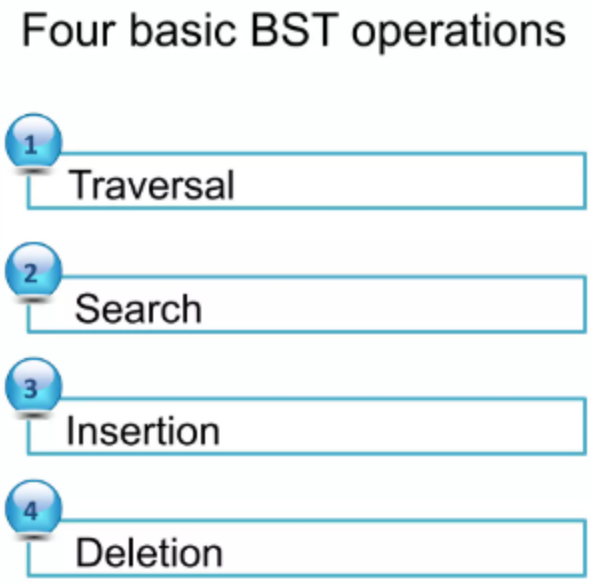


Difference between Binary Tree and Binary Search Tree

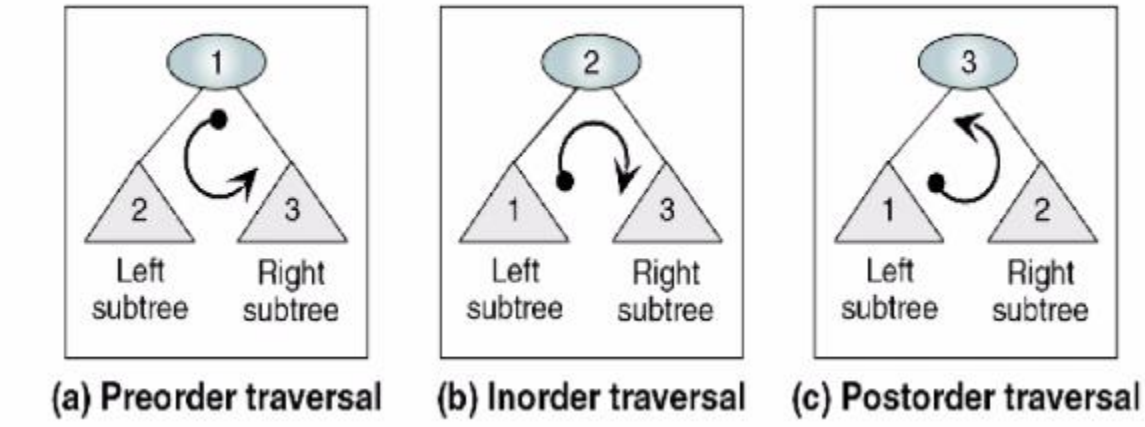
A binary tree is simply a tree in which each node can have at most two children.

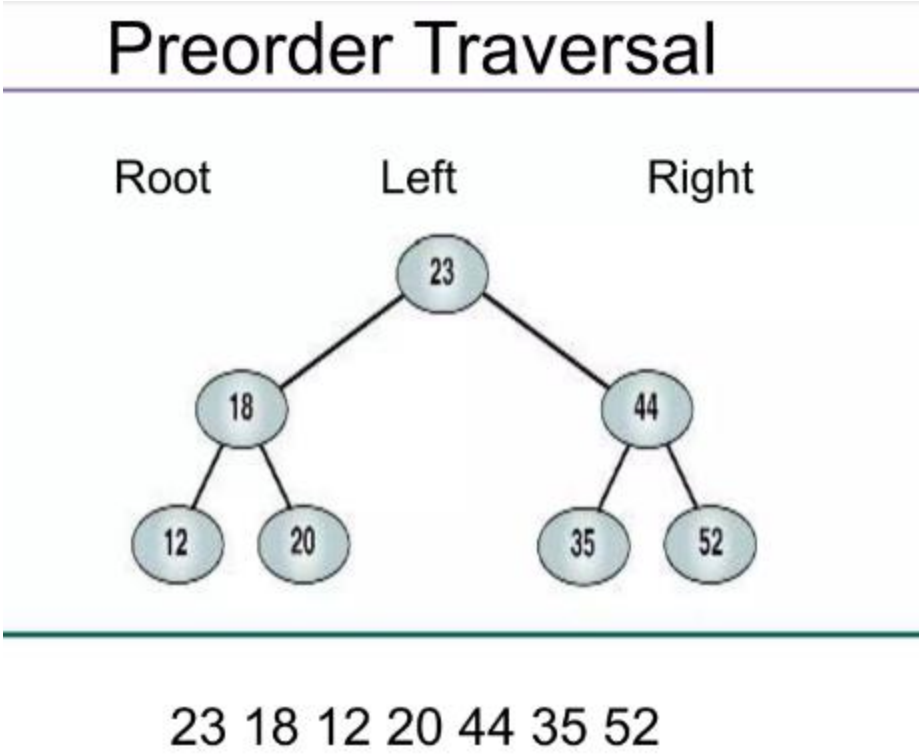
A binary search tree is a binary tree in which the nodes are assigned values, with the following restrictions :

1. No duplicate values.
2. The left subtree of a node can only have values less than the node
3. The right subtree of a node can only have values greater than the node and recursively defined
4. The left subtree of a node is a binary search tree.
5. The right subtree of a node is a binary search tree.



BST Traversal





Algorithm PreOrderTraversal(BSTNode root)

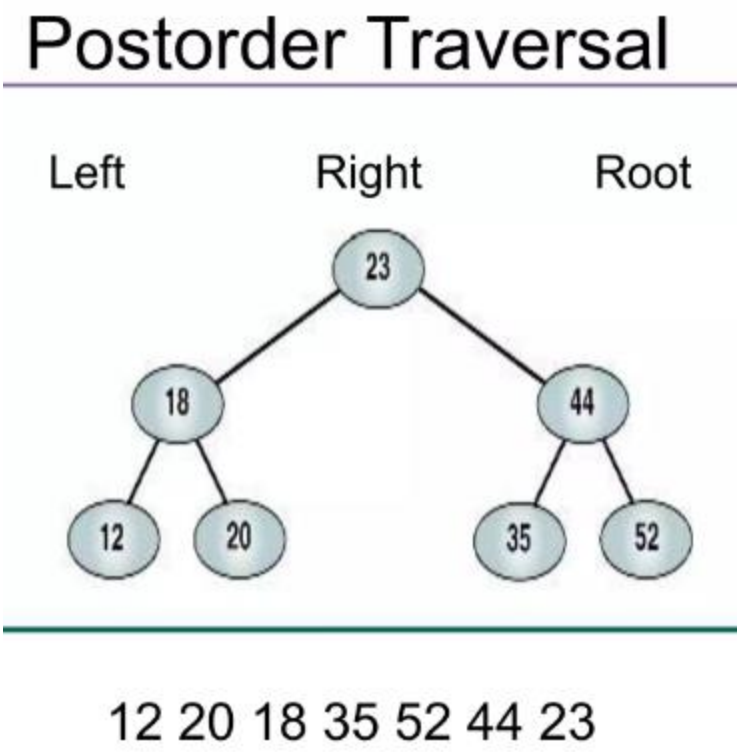
1. if root == NULL

2. return

3. Print root->value (or perform any operation on the value)

4. PreOrderTraversal(root->left)

5. PreOrderTraversal(root->right)



Algorithm PostOrderTraversal(BSTNode root)

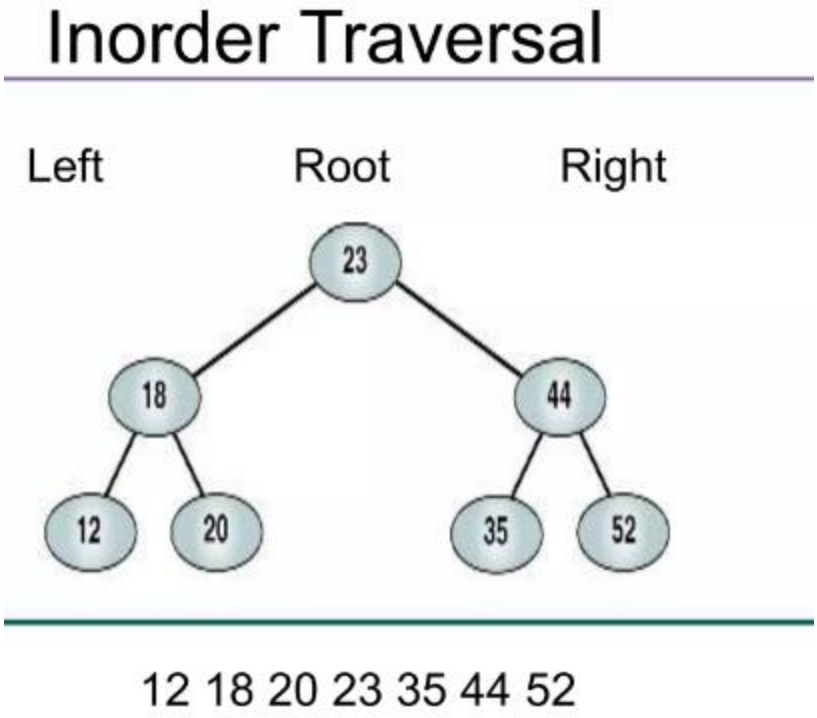
1. if root == NULL

2. return

3. PostOrderTraversal(root->left)

4. PostOrderTraversal(root->right)

5. Print root->value (or perform any operation on the value)



Algorithm InOrderTraversal(BSTNode root)

1. if root == NULL

2. return

3. InOrderTraversal(root->left)

4. Print root->value (or perform any operation on the value)

5. InOrderTraversal(root->right)